



Chittagong Independent University
School of Science & Engineering
Department of Computer Science and Engineering (CSE)

Project Report
2D Animated Journey

Course: Computer Graphics Lab
Course Code: CSE 435L

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Abstract

This project presents the design and implementation of a 2D animated short scene illustrating the daily journey of a Chittagong Independent University (CIU) student from Jamalkhan bus stop to campus life and back. The animation aims to combine realistic 2D modeling, detailed texturing, and smooth character animation to depict various stages of student's day—arriving at Jamalkhan, walking to the CIU campus, attending a class, participating in a seminar, eating at the cafeteria, and returning home. The project uses C++ with OpenGL for rendering and for modeling, incorporating advanced techniques such as keyframe animation, camera transitions, and dynamic environmental effects. The final product delivers an immersive experience that blends technical graphics knowledge with a relatable narrative, demonstrating both academic and practical applications in campus visualization and digital storytelling.

Chapter-01

Introduction

Background Information

The evolution of computer graphics technologies has made it increasingly possible to produce interactive and highly realistic 2D environments. These technologies are widely used in entertainment, education, urban planning, and promotional media. For universities, they offer a way to showcase campus life in an engaging and visually immersive manner. A well-crafted 3D animation can go beyond static images or videos by creating a dynamic, story-driven experience.

This project aims to leverage 2D modeling, animation, and cinematic camera techniques to create a short animated scene that visually represents the daily life of a Chittagong Independent University (CIU) student.

Problem Statement

Most existing campus-related animations tend to be generic, template-based, and location-independent, resulting in minimal emotional connection with the intended audience. They often fail to capture the unique atmosphere, culture, and landmarks of a specific location. This lack of contextual realism reduces engagement and diminishes the storytelling potential.

To address this gap, this project will produce a location-specific, story-driven 2D animation focused on CIU and its surrounding Jamalkhan area. The animation will depict real campus landmarks and relatable daily student activities to increase immersion and relevance.

Objectives

1. Create realistic 2D models of Jamalkhan bus stop and key CIU campus landmarks including classrooms, seminar hall, and cafeteria.
2. Implement smooth animations for the student character and surrounding environmental elements such as vehicles, trees, and pedestrians.
3. Integrate cinematic camera movements such as pans, zooms, and transitions to enhance storytelling and maintain viewer engagement.

Scope

The project covers the complete pipeline of 2D animation production, including:

- **Modeling:**Detailed creation of CIU and Jamalkhan landmarks.
- **Texturing:**Applying realistic surface materials for buildings, vehicles, and characters.
- **Lighting:**Simulating daylight effects and shadows for realism.
- **Animation:**Character movement, environmental dynamics, and camera transitions.
- **Rendering:**Outputting scenes in high quality for playback.
- **Scene Composition:**Arranging visual elements for cinematic appeal across multiple locations in Jamalkhan and CIU.

Importance of the Project

This project holds significance both academically and practically:

- **Academic Value:**Strengthens understanding of advanced computer graphics techniques such as hierarchical modeling,keyframe animation,and camera path planning.
- **Practical Value:**Produces a location-specific promotional animation that can be used by CIU for orientation,marketing,or virtual campus tours.
- **Engagement:**By accurately depicting recognizable locations and real-life scenarios,it creates an emotional connection with viewers—especially prospective students and alumni.

Chapter-02

Literature Review / Background Study

2D campus visualizations have been increasingly adopted by universities and educational institutions as a means to engage students, promote facilities, and provide virtual tours. These visualizations offer significant benefits in orientation programs, helping new students become familiar with the layout and facilities of a campus before physically arriving.

In previous works, scene graph modeling has been used to efficiently manage and render complex 2D environments by organizing them hierarchically. Texture mapping has enhanced realism by applying high-quality images to 3D surfaces, while keyframe animation has brought characters and environmental elements to life through interpolated motion between defined states.

This project draws inspiration from both educational and commercial 2D animation techniques:

- Scene Graph Modeling-enables the logical structuring of scenes where buildings, vehicles, characters, and props are grouped and transformed together.
- Texture Mapping-provides lifelike surfaces for buildings, roads, and props by using photographic textures captured from real Jamalkhan and CIU locations.
- Keyframe Animation-ensures fluid movements for characters(walking, sitting, eating) and objects(moving buses, waving trees).

This project specifically builds upon the following technologies and practices:

1. OpenGL Rendering Pipeline–For real-time rendering of 2D models and animations, including shading, transformation, and rasterization processes.
2. Blender Modeling Workflows–For asset creation, including polygon modeling, UV unwrapping for textures, and rigging for animation.
3. Hierarchical Animation–For managing complex articulated characters, ensuring that movements of limbs, head, and body are naturally coordinated.

By combining these established techniques with accurate real-world location modeling, this project aims to produce a location-specific animated short film that not only entertains but also informs and connects with its audience—particularly current and prospective CIU students.

Chapter-03

Methodology / Project Design

Tools & Technologies

- Programming Language: C++—Chosen for its performance and strong integration with graphics APIs.
- Graphics Library: OpenGL (with GLUT)—Used for cross-platform rendering, handling input, and managing the rendering loop.
- IDE: Code::Blocks—Selected for efficient C++ development with OpenGL integration.
- Platform: Windows 10—Target platform for development and deployment.

Design Steps

1. Environment Modeling
 - Create 2D models of Jamalkhan bus stop, CIU main gate, classrooms, seminar hall, cafeteria, and roads.
 - Ensure scale accuracy to match real-world proportions.
 - Model environmental props such as streetlights, benches, signboards, and trees for added realism.
2. Character Modeling
 - Design a 2D student character with appropriate attire to reflect CIU's student life.
 - Apply rigging for skeletal animation, enabling body parts like arms, legs, and head to move independently.
 - Prepare variations for facial expressions to enhance storytelling.
3. Animation
 - Implement keyframe animations for walking, sitting, standing, eating, writing, and interacting with the environment.
 - Animate environmental elements such as moving buses, cars, clouds, and swaying trees for liveliness.
 - Ensure motion smoothing using interpolation techniques.
4. Camera Movement
 - Design cinematic camera paths to follow the student's journey.
 - Use first-person perspective during walking segments for immersion and third-person perspective for environmental overviews.
 - Integrate panning, tracking, and zooming to highlight key moments (e.g., entering the CIU gate, attending class).

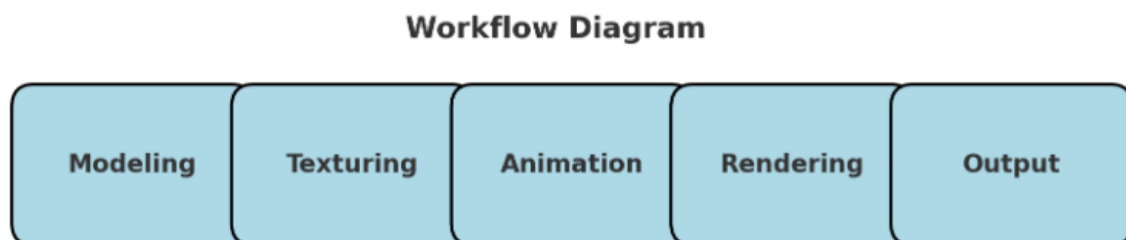
5. Lighting & Texturing

- Apply daylight simulation to mimic real-world outdoor lighting.
- Use texture maps from photographs of CIU buildings and Jamalkhan area for authenticity.
- Add shadow mapping to enhance depth and realism in scenes.

6. Rendering

- Use the OpenGL rendering pipeline for real-time display of all assets and animations.
- Optimize models using Level of Detail (LOD) techniques to ensure smooth performance without sacrificing quality.
- Implement double buffering to avoid flicker during animation playback.

Diagram of my method:



Chapter-04

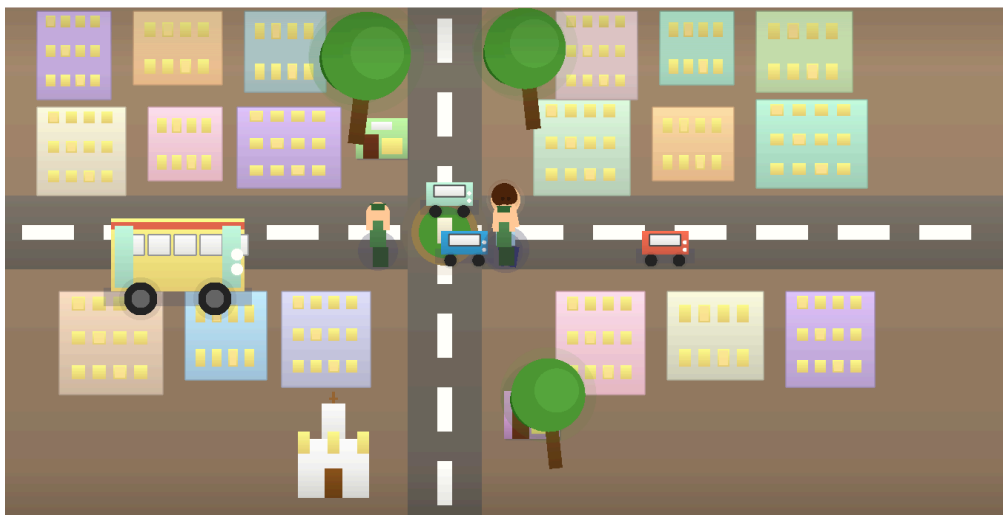
Implementation & Results

Scenes

- Scene 0(Jamalkhan Bus Stop):Student steps off a bus with animated vehicles and pedestrians.
- Scene 1(Senior Citizens Club Road)-Smooth walking animation with swaying trees and buildings,sidewalks with vendors,traffic moving.
- Scene 2(Uphill to CIU):Smooth walking animation with swaying trees and buildings.
- Scene 3(CIU Building)-CIU building with main gate,cars parked,builings on both side.
- Scene 4(Cafeteria):Student eating lunch with friends.
- Scene 5(Computer lab):Student attending class in computer lab.
- Scene 6(Seminar Hall):Guest speaker,faculties and audience reactions.
- Scene 7(Library):Library officer and students studying.
- Scene 8(Return):Student walks back to Jamalkhan to take the bus home.

Output Screenshots:

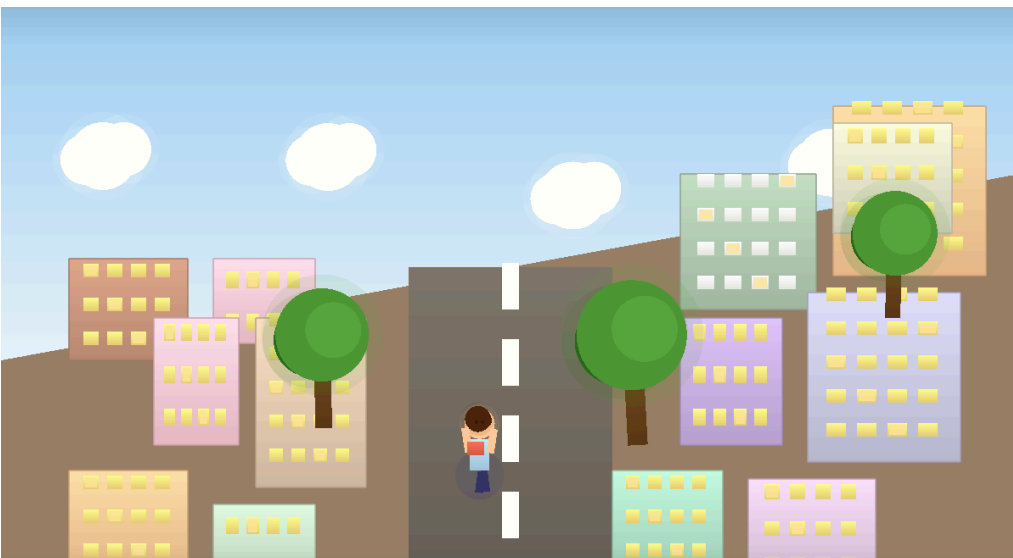
1.Jamalkhan Bus Stop



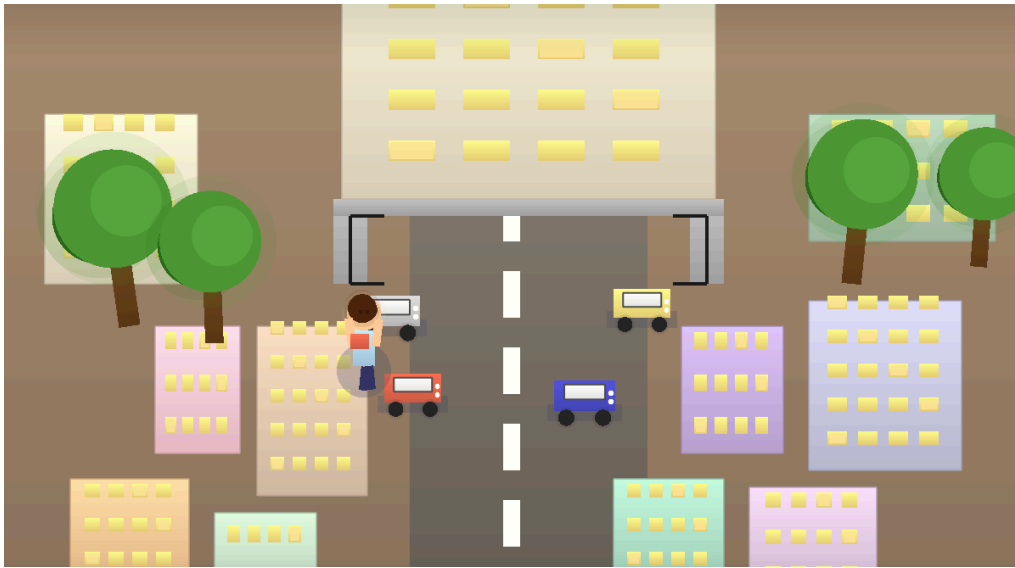
2.Senior Citizens Club Road



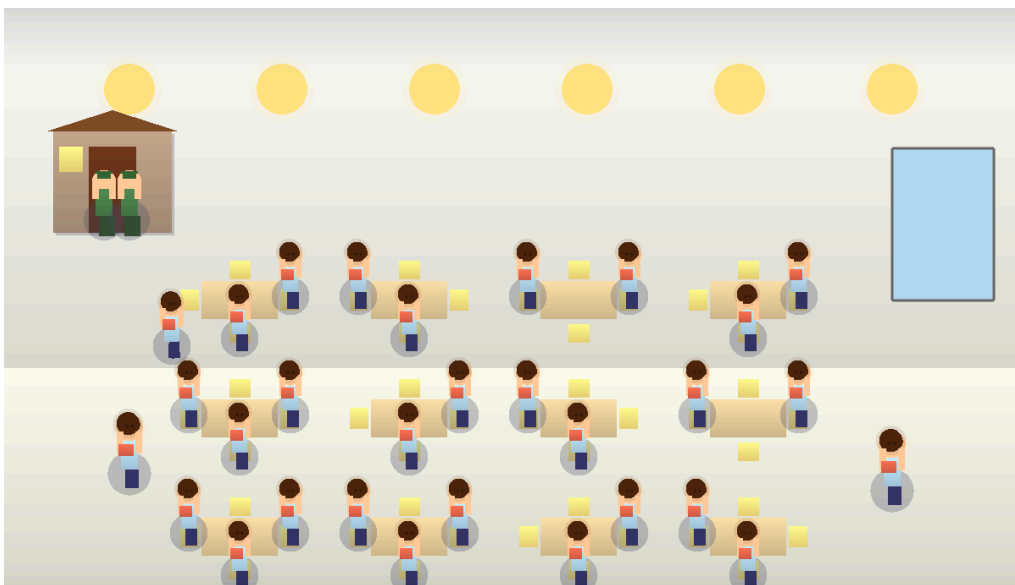
3.Uphill to CIU



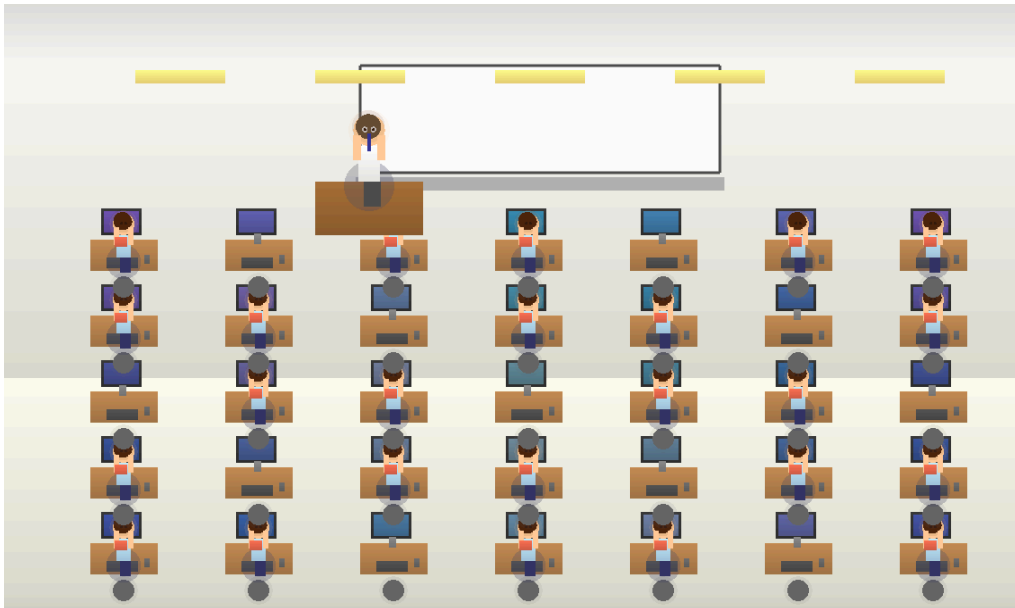
4.CIU Building



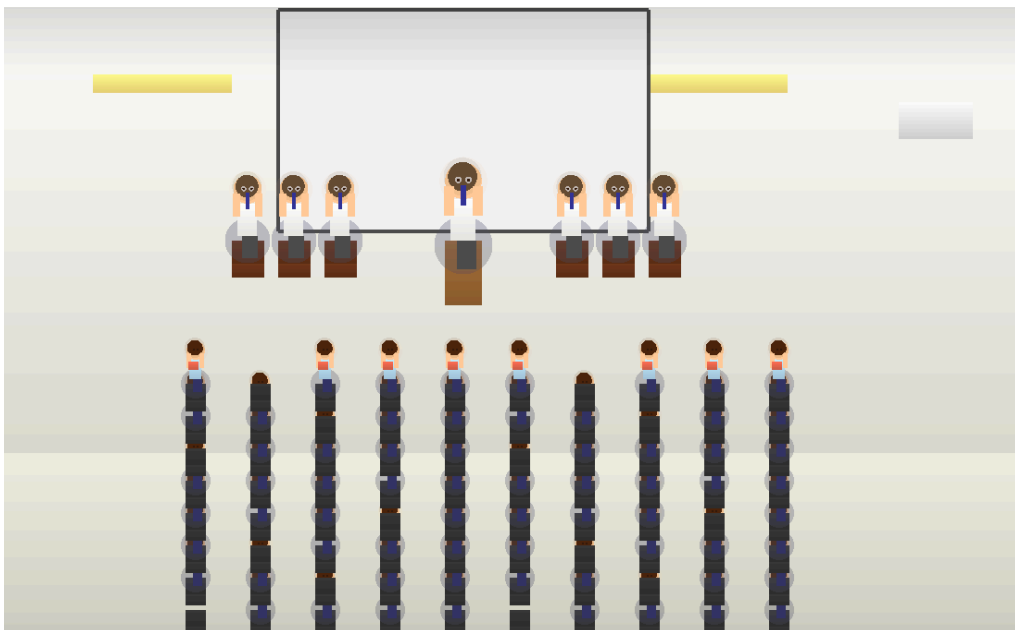
5.Cafeteria



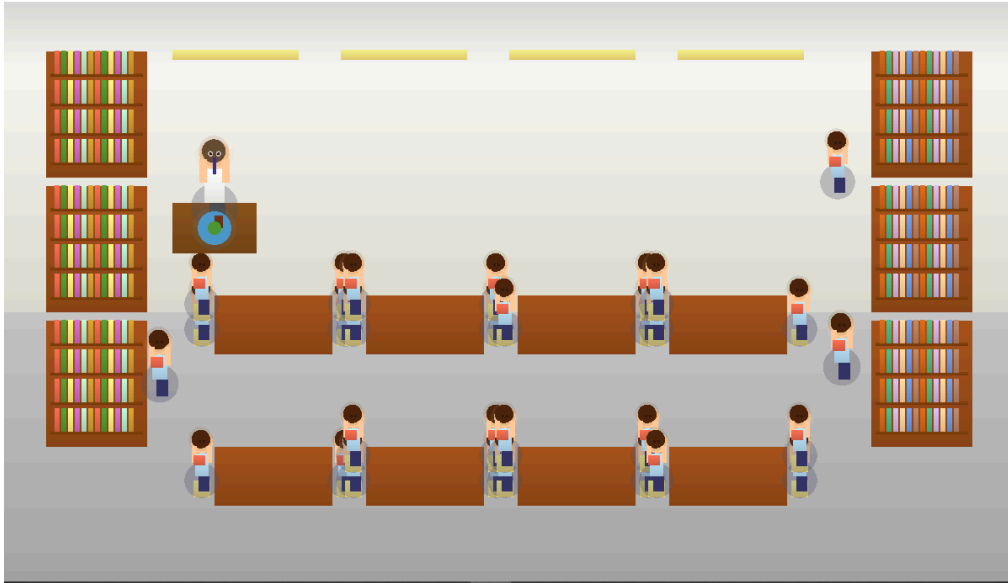
6. Computer Lab



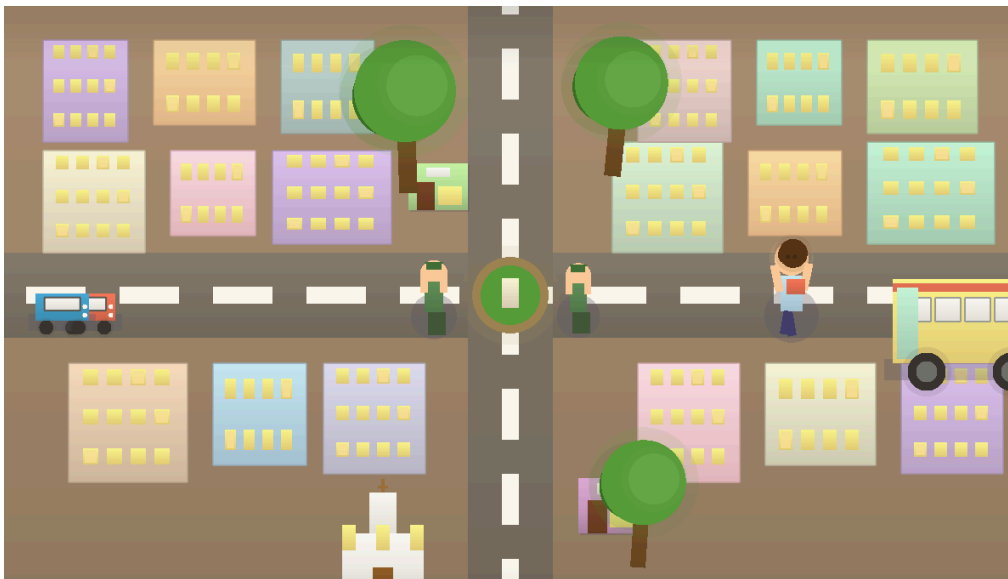
7. Seminar Hall



8.Enhanced Library



9.Return to Jamalkhan Bus Stop



Chapter-05

Discussion / Analysis

The developed animation effectively captures the essence of a CIU student's journey from Jamalkhan to the campus and back, with notable attention to location-specific details. Key features such as the CIU main gate, recognizable buildings, bus stop, and cafeteria were modeled to reflect their real-world counterparts, creating a sense of familiarity for anyone acquainted with the location.

The use of multiple camera angles—including first-person, third-person, panning shots, and zoom-ins—adds cinematic variety and keeps the audience engaged throughout the film. These camera movements were synchronized with scene transitions to maintain narrative flow and visual appeal.

Dynamic lighting was applied to mimic real-world daylight, with shadows shifting naturally based on camera positioning and environmental geometry. This greatly enhanced depth perception and realism, especially during outdoor walking scenes.

Performance testing showed consistently smooth frame rates across all scenes, indicating that the optimization of polygon counts and use of Level of Detail (LOD) techniques were successful in balancing visual quality and rendering speed.

Despite the overall success, the project encountered several technical challenges:

- **Model Complexity Optimization:** High-detail assets had to be simplified to ensure they could be rendered in real time without significant frame drops.
- **Natural Character Motion:** Achieving realistic walking cycles required iterative adjustments to joint rotations, stride lengths, and limb synchronization.
- **Scene Transitions:** Coordinating seamless transitions between multiple environments while maintaining camera continuity demanded precise timeline and animation curve control.

These challenges were addressed through iterative testing, feedback loops, and targeted optimization, resulting in a stable and visually engaging animation.

Chapter-06

Conclusion & Future Work

The project successfully achieved its primary objective of creating a 3D animated scene that depicts the daily routine of a CIU student with a blend of technical accuracy and creative storytelling.

- **Technical Achievement:** The animation pipeline incorporated advanced modeling, texturing, lighting, and camera control techniques, executed using Blender for asset creation and OpenGL for rendering.
- **Narrative Strength:** The storyline was relatable and location-specific, making it emotionally engaging for the intended audience.
- **Visual Quality:** Realistic textures, dynamic lighting, and carefully choreographed camera work produced an immersive viewing experience.

Overall, the animation demonstrates how computer graphics can be used not only as a technical showcase but also as a storytelling medium that resonates with its audience.

Future Work

The project opens several possibilities for enhancement and expansion:

1. **Virtual Reality (VR) Integration** – Transform the animation into an interactive VR campus tour where users can freely navigate environments in first person.
2. **3D Implementation** – Make 3D structure to guide viewers through scenes, complemented by ambient sound effects and background music to enhance immersion.
3. **User-Driven Navigation** – Develop an interactive version where viewers can choose their own paths (e.g., visit the library instead of cafeteria) using a menu-based navigation system.
4. **Higher-Detail Assets for Cinematic Rendering** – Use ultra-high-resolution textures and detailed models for offline rendering in promotional videos.
5. **Weather & Time Variations** – Introduce environmental variations such as rainy days, evening lighting, and seasonal changes to increase replay value and realism.
6. **Multi-Character Interaction** – Expand the scenes to include more animated students, staff, and activities to portray a busier campus environment.

By implementing these improvements, the project could evolve from a fixed short film into a fully interactive and immersive campus experience, bridging the gap between entertainment, education, and promotional media.

References

- Angel, E., & Shreiner, D. (2019). *Interactive computer graphics*. Pearson.
- Feiner, S. K., & Van Dam, A. (2006). *Schaum's outline of computer graphics* (2nd ed.). McGraw-Hill Education.
- OpenGL. (n.d.). *OpenGL documentation*. <https://www.opengl.org>
- Blender Foundation. (n.d.). *Blender manual*. <https://docs.blender.org>
- OpenAI. (n.d.). *ChatGPT*. <https://chatgpt.com/>
- Anthropic. (n.d.). *Claude*. <https://claude.ai/>

Appendices

Appendix A – Source Code

```
// Scene 0: Jamalkhan Bus Stop with buildings on all sides
void drawEnhancedJamalkhanBusStop() {
    drawEnhancedSky();

    // Ground level
    drawRect(0, 0, windowWidth, windowHeight, GROUND_COLOR, true);

    // Main horizontal road
    drawRect(0, 350, windowWidth, 100, GHIBLI_ROAD, true);

    // Vertical intersection road
    drawRect(550, 0, 100, windowHeight, GHIBLI_ROAD, true);

    // Road markings - horizontal
    setColor(GHIBLI_WHITE);
    for (int i = 0; i < 12; i++) {
        drawRect(i * 110 + 30, 390, 70, 20, GHIBLI_WHITE);
    }

    // Road markings - vertical
    for (int i = 0; i < 8; i++) {
        drawRect(590, i * 100 + 30, 20, 60, GHIBLI_WHITE);
    }

    // Central roundabout
    drawCircle(600, 400, 45, Color(0.6f, 0.5f, 0.3f), true);
    drawCircle(600, 400, 35, GHIBLI_TREE_GREEN);
```



```

// Monument in roundabout
drawRect(590, 385, 20, 35, GHIBLI_CREAM, true);

// Buildings on all four sides - properly placed on sidewalks
// Top side buildings
drawEnhancedBuilding(50, 450, 120, 120, GHIBLI_CREAM,
GHIBLI_WARM_YELLOW, 3);
drawEnhancedBuilding(200, 470, 100, 100, GHIBLI_WARM_PINK,
GHIBLI_WARM_YELLOW, 2);
drawEnhancedBuilding(320, 460, 140, 110, GHIBLI_LAVENDER,
GHIBLI_WARM_YELLOW, 3);
drawEnhancedBuilding(720, 450, 130, 130, Color(0.8f, 0.9f, 0.8f),
GHIBLI_WARM_YELLOW, 3);
drawEnhancedBuilding(880, 470, 110, 100, GHIBLI_PEACH,
GHIBLI_WARM_YELLOW, 2);
drawEnhancedBuilding(1020, 460, 150, 120, GHIBLI_MINT,
GHIBLI_WARM_YELLOW, 3);

// Bottom side buildings
drawEnhancedBuilding(80, 180, 140, 140, Color(0.9f, 0.8f, 0.7f),
GHIBLI_WARM_YELLOW, 3);
drawEnhancedBuilding(250, 200, 110, 120, GHIBLI_SOFT_BLUE,
GHIBLI_WARM_YELLOW, 2);
drawEnhancedBuilding(380, 190, 120, 130, Color(0.8f, 0.8f, 0.9f),
GHIBLI_WARM_YELLOW, 3);
drawEnhancedBuilding(750, 180, 120, 140, GHIBLI_WARM_PINK,
GHIBLI_WARM_YELLOW, 3);
drawEnhancedBuilding(900, 200, 130, 120, Color(0.9f, 0.9f, 0.8f),
GHIBLI_WARM_YELLOW, 2);
drawEnhancedBuilding(1060, 190, 120, 130, GHIBLI_LAVENDER,
GHIBLI_WARM_YELLOW, 3);

// Left side buildings
drawEnhancedBuilding(50, 580, 100, 120, Color(0.8f, 0.7f, 0.9f),
GHIBLI_WARM_YELLOW, 3);
drawEnhancedBuilding(180, 600, 120, 100, GHIBLI_PEACH,
GHIBLI_WARM_YELLOW, 2);
drawEnhancedBuilding(330, 590, 110, 110, Color(0.7f, 0.8f, 0.8f),
GHIBLI_WARM_YELLOW, 2);

// Right side buildings
drawEnhancedBuilding(750, 580, 110, 120, Color(0.9f, 0.8f, 0.8f),
GHIBLI_WARM_YELLOW, 3);

```

```

drawEnhancedBuilding(890, 600, 100, 100, GHIBLI_MINT, GHIBLI_WARM_YELLOW,
2);
drawEnhancedBuilding(1020, 590, 130, 110, Color(0.8f, 0.9f, 0.7f),
GHIBLI_WARM_YELLOW, 2);

// Church - made more visible
drawChurch(450, 40, 0.8f);

// Shops on sidewalks
drawShop(480, 500, 70, 55, Color(0.7f, 0.9f, 0.6f), "general");
drawShop(680, 120, 75, 65, Color(0.8f, 0.6f, 0.8f), "general");

// Trees on sidewalks
drawEnhancedTree(480, 520, 1.2f);
drawEnhancedTree(720, 540, 1.1f);
drawEnhancedTree(750, 80, 1.0f);

// Enhanced bus
float currentBusX = -250 + animationProgress * 0.2f;
if (currentBusX > 150) currentBusX = 150;
drawEnhancedBus(currentBusX, 320, 0.9f);

// Traffic on roads
float car1X = fmod(animationProgress * 0.4f, windowWidth + 300) - 150;
float car2X = fmod(animationProgress * 0.3f + 400, windowWidth + 300) - 150;

if (car1X > -120 && car1X < windowWidth) {
    drawEnhancedCar(car1X, 370, GHIBLI_SOFT_RED);
}
if (car2X > -120 && car2X < windowWidth) {
    drawEnhancedCar(car2X, 370, Color(0.2f, 0.6f, 0.8f));
}

// Vertical road traffic
float carVertical1Y = fmod(animationProgress * 0.35f, windowHeight + 200) - 100;
if (carVertical1Y > -80 && carVertical1Y < windowHeight) {
    drawEnhancedCar(575, carVertical1Y, GHIBLI_MINT);
}

// Students and pedestrians
if (animationProgress > 1500) {
    float student1X = currentBusX + 220 + (animationProgress - 1500) * 0.06f;
    drawEnhancedStudent(student1X, 380, true);
}

```

```

    }

    // Guards
    drawGuard(510, 380, 0.9f);
    drawGuard(680, 380, 0.85f);
}

// Scene 1: Senior Citizens Club - removed sky, more ground buildings
void drawEnhancedSeniorCitizensClub() {
    // All ground - no sky
    drawRect(0, 0, windowWidth, windowHeight, GROUND_COLOR, true);

    // Main vertical road
    drawRect(500, 0, 200, windowHeight, GHIBLI_ROAD, true);

    // Road markings
    setColor(GHIBLI_WHITE);
    for (int i = 0; i < 10; i++) {
        drawRect(590, i * 90, 20, 55, GHIBLI_WHITE);
    }

    // Vertical sidewalks
    drawVerticalSidewalk(500, 0, windowHeight, false);
    drawVerticalSidewalk(700, 0, windowHeight, true);

    // Left side - Government school and other buildings
    drawGovtSchool(200, 450, 0.8f);
    drawEnhancedBuilding(50, 400, 120, 180, Color(0.8f, 0.7f, 0.6f),
GHIBLI_WARM_YELLOW, 4);
    drawEnhancedBuilding(180, 200, 100, 150, GHIBLI_WARM_PINK,
GHIBLI_WARM_YELLOW, 3);
    drawEnhancedBuilding(300, 150, 130, 200, Color(0.9f, 0.8f, 0.7f),
GHIBLI_WARM_YELLOW, 5);
    drawEnhancedBuilding(80, 50, 140, 120, GHIBLI_PEACH, GHIBLI_WARM_YELLOW,
3);
    drawEnhancedBuilding(250, 30, 120, 100, Color(0.8f, 0.9f, 0.8f),
GHIBLI_WARM_YELLOW, 2);

    // Right side - Senior Citizens Club and other buildings
    drawEnhancedBuilding(750, 400, 200, 180, GHIBLI_WHITE,
GHIBLI_WARM_YELLOW, 4); // Main Senior Citizens Club
    drawEnhancedBuilding(980, 450, 140, 130, Color(0.9f, 0.9f, 0.8f),
GHIBLI_WARM_YELLOW, 3);

```

```

    drawEnhancedBuilding(800, 200, 120, 150, GHIBLI_LAVENDER,
GHIBLI_WARM_YELLOW, 3);
    drawEnhancedBuilding(950, 180, 180, 200, Color(0.8f, 0.8f, 0.9f),
GHIBLI_WARM_YELLOW, 5);
    drawEnhancedBuilding(720, 50, 130, 120, GHIBLI_MINT, GHIBLI_WARM_YELLOW,
3);
    drawEnhancedBuilding(880, 30, 150, 130, Color(0.9f, 0.8f, 0.9f),
GHIBLI_WARM_YELLOW, 3);

```

```

// Shops along sidewalks

```

```

drawShop(450, 520, 70, 50, Color(0.8f, 0.6f, 0.7f), "general");
drawShop(430, 350, 65, 45, Color(0.7f, 0.8f, 0.6f), "food");
drawShop(440, 200, 75, 55, Color(0.6f, 0.7f, 0.9f), "general");
drawShop(720, 520, 70, 50, Color(0.9f, 0.7f, 0.6f), "food");
drawShop(730, 350, 65, 45, Color(0.8f, 0.8f, 0.6f), "general");
drawShop(740, 180, 75, 55, Color(0.7f, 0.6f, 0.8f), "general");

```

```

// Trees scattered around

```

```

drawEnhancedTree(120, 350, 1.4f);
drawEnhancedTree(350, 380, 1.2f);
drawEnhancedTree(450, 450, 1.1f);
drawEnhancedTree(780, 350, 1.3f);
drawEnhancedTree(950, 320, 1.0f);
drawEnhancedTree(1100, 400, 1.2f);

```

```

// Enhanced vehicles on road

```

```

float bus1Y = fmod(animationProgress * 0.35f, windowHeight + 300) - 150;
float car1Y = fmod(animationProgress * 0.4f + 200, windowHeight + 300) - 150;

```

```

if (bus1Y > -100 && bus1Y < windowHeight) {
    drawRect(530, bus1Y, 90, 55, Color(0.1f, 0.8f, 0.2f), true); // CNG
    drawCircle(550, bus1Y - 15, 16, Color(0.1f, 0.1f, 0.1f), true);
    drawCircle(600, bus1Y - 15, 16, Color(0.1f, 0.1f, 0.1f), true);
}

```

```

if (car1Y > -80 && car1Y < windowHeight) {
    drawEnhancedBus(520, car1Y, 0.6f);
}

```

```

// Multiple students and pedestrians

```

```

float student1Y = fmod(animationProgress * 0.08f, windowHeight + 100) - 50;
float student2Y = fmod(animationProgress * 0.06f + 300, windowHeight + 100) - 50;

```

```

if (student1Y > 0 && student1Y < windowHeight) {

```

```

        drawEnhancedStudent(470, student1Y, true, 0.9f);
    }
    if (student2Y > 0 && student2Y < windowHeight) {
        drawEnhancedStudent(750, student2Y, true, 0.85f, -1);
    }

    // Guards near important buildings
    drawGuard(480, 450, 0.9f);
    drawGuard(720, 380, 0.85f);
}

// Scene 2: Enhanced Uphill to CIU
void drawEnhancedUphillToCIU() {
    drawEnhancedSky();

    // Sloped ground
    setColor(GROUND_COLOR);
    glBegin(GL_QUADS);
    glVertex2f(0, 0);
    glVertex2f(windowWidth, 0);
    glVertex2f(windowWidth, 520);
    glVertex2f(0, 300);
    glEnd();

    // Main uphill road
    drawRect(480, 40, 240, 370, GIBLI_ROAD, true);

    // Road markings
    setColor(GIBLI_WHITE);
    for (int i = 0; i < 5; i++) {
        drawRect(590, i * 90, 20, 55, GIBLI_WHITE);
    }

    // Buildings on sidewalks properly placed
    drawEnhancedBuilding(80, 300, 140, 120, Color(0.8f, 0.6f, 0.5f),
GIBLI_WARM_YELLOW, 3);
    drawEnhancedBuilding(250, 320, 120, 100, GIBLI_WARM_PINK,
GIBLI_WARM_YELLOW, 2);
    drawEnhancedBuilding(800, 360, 160, 160, Color(0.7f, 0.8f, 0.7f), GIBLI_WHITE, 4);
    drawEnhancedBuilding(980, 400, 180, 200, GIBLI_PEACH,
GIBLI_WARM_YELLOW, 5);
    drawEnhancedBuilding(980, 450, 140, 130, Color(0.9f, 0.9f, 0.8f),
GIBLI_WARM_YELLOW, 3);

```

```

drawEnhancedBuilding(800, 200, 120, 150, GHIBLI_LAVENDER,
GHIBLI_WARM_YELLOW, 3);
    drawEnhancedBuilding(950, 180, 180, 200, Color(0.8f, 0.8f, 0.9f),
GHIBLI_WARM_YELLOW, 5);
    drawEnhancedBuilding(720, 50, 130, 120, GHIBLI_MINT, GHIBLI_WARM_YELLOW,
3);
    drawEnhancedBuilding(880, 30, 150, 130, Color(0.9f, 0.8f, 0.9f),
GHIBLI_WARM_YELLOW, 3);
    drawEnhancedBuilding(180, 200, 100, 150, GHIBLI_WARM_PINK,
GHIBLI_WARM_YELLOW, 3);
    drawEnhancedBuilding(300, 150, 130, 200, Color(0.9f, 0.8f, 0.7f),
GHIBLI_WARM_YELLOW, 5);
    drawEnhancedBuilding(80, 50, 140, 120, GHIBLI_PEACH, GHIBLI_WARM_YELLOW,
3);
    drawEnhancedBuilding(250, 30, 120, 100, Color(0.8f, 0.9f, 0.8f),
GHIBLI_WARM_YELLOW, 2);

// Trees on slope
drawEnhancedTree(380, 220, 1.1f);
drawEnhancedTree(750, 200, 1.3f);
drawEnhancedTree(1050, 350, 1.0f);

// Student walking uphill
float slopeProgress = fmod(animationProgress * 0.05f, 300);
float studentSlopeX = 520 + slopeProgress * 0.4f;
float studentSlopeY = 80 + slopeProgress * 0.8f;
drawEnhancedStudent(studentSlopeX, studentSlopeY, true, 0.9f);
}

// Scene 3: Enhanced CIU Building - fixed placement
void drawEnhancedCIUBuilding() {
    drawEnhancedSky();

    // Ground level
    drawRect(0, 0, windowWidth, windowHeight, GROUND_COLOR, true);

    // Main uphill road
    drawRect(480, 40, 280, 440, GHIBLI_ROAD, true);

    // Road markings
    setColor(GHIBLI_WHITE);
    for (int i = 0; i < 6; i++) {
        drawRect(590, i * 90, 20, 55, GHIBLI_WHITE);
    }
}

```

```

// Enhanced CIU gate properly on ground
drawEnhancedCIUGate(450, 400);

// Main CIU building properly grounded behind gate
drawEnhancedBuilding(400, 500, 440, 300, GHIBLI_CREAM,
GHIBLI_WARM_YELLOW, 5);

// Parked cars beside road (not on road)
drawEnhancedCar(420, 350, Color(0.8f, 0.8f, 0.8f), 0.8f);
drawEnhancedCar(450, 260, GHIBLI_SOFT_RED, 0.75f);
drawEnhancedCar(650, 250, Color(0.3f, 0.3f, 0.8f), 0.8f);
drawEnhancedCar(720, 360, GHIBLI_WARM_YELLOW, 0.75f);

// Surrounding buildings on proper ground
drawEnhancedBuilding(50, 400, 180, 200, GHIBLI_CREAM,
GHIBLI_WARM_YELLOW, 4);
drawEnhancedBuilding(950, 450, 220, 150, Color(0.7f, 0.8f, 0.7f),
GHIBLI_WARM_YELLOW, 3);
drawEnhancedBuilding(180, 200, 100, 150, GHIBLI_WARM_PINK,
GHIBLI_WARM_YELLOW, 3);
drawEnhancedBuilding(300, 150, 130, 200, Color(0.9f, 0.8f, 0.7f),
GHIBLI_WARM_YELLOW, 5);
drawEnhancedBuilding(80, 50, 140, 120, GHIBLI_PEACH, GHIBLI_WARM_YELLOW,
3);
drawEnhancedBuilding(250, 30, 120, 100, Color(0.8f, 0.9f, 0.8f),
GHIBLI_WARM_YELLOW, 2);
drawEnhancedBuilding(800, 200, 120, 150, GHIBLI_LAVENDER,
GHIBLI_WARM_YELLOW, 3);
drawEnhancedBuilding(950, 180, 180, 200, Color(0.8f, 0.8f, 0.9f),
GHIBLI_WARM_YELLOW, 5);
drawEnhancedBuilding(720, 50, 130, 120, GHIBLI_MINT, GHIBLI_WARM_YELLOW,
3);
drawEnhancedBuilding(880, 30, 150, 130, Color(0.9f, 0.8f, 0.9f),
GHIBLI_WARM_YELLOW, 3);

// Trees around campus
drawEnhancedTree(150, 350, 1.4f);
drawEnhancedTree(250, 330, 1.2f);
drawEnhancedTree(1000, 400, 1.3f);
drawEnhancedTree(1150, 420, 1.1f);

// Students approaching
float approachProgress1 = fmod(animationProgress * 0.04f, 200);

```

```

drawEnhancedStudent(250 + approachProgress1, 300, true);

}

// Scene 4: Enhanced CIU Cafeteria
void drawEnhancedCIUCafeteria() {
    // Warm floor
    drawRect(0, 0, windowWidth, 320, Color(0.9f, 0.9f, 0.85f), true);

    // Soft walls
    drawRect(0, 320, windowWidth, windowHeight - 320, Color(0.95f, 0.95f, 0.9f), true);

    // Ceiling lights
    for (int i = 0; i < 6; i++) {
        float lightX = 150 + i * 180;
        drawCircle(lightX, 650, 30, GHIBLI_WARM_YELLOW, true);
    }

    // Guard room
    drawEnhancedGuardRoom(60, 480, 140, 120, Color(0.7f, 0.6f, 0.5f));
    drawGuard(120, 500, 0.8f);
    drawGuard(150, 500, 0.8f);

    // Tables and chairs with more students
    for (int row = 0; row < 3; row++) {
        for (int col = 0; col < 4; col++) {
            float tableX = 280 + col * 200;
            float tableY = 120 + row * 140;

            // Table
            drawRect(tableX - 45, tableY - 22, 90, 45, Color(0.9f, 0.8f, 0.6f), true);

            // Yellow chairs and students
            Color chairColor = GHIBLI_WARM_YELLOW;
            drawRect(tableX - 70, tableY - 12, 22, 25, chairColor, true);
            drawRect(tableX + 48, tableY - 12, 22, 25, chairColor, true);
            drawRect(tableX - 12, tableY - 50, 25, 22, chairColor, true);
            drawRect(tableX - 12, tableY + 25, 25, 22, chairColor, true);

            // Students at tables
            if ((row + col) % 3 != 2) {
                drawEnhancedStudent(tableX - 60, tableY + 10, false, 0.7f);
            }
        }
    }
}

```



```

        if ((row + col) % 4 != 3) {
            drawEnhancedStudent(tableX + 60, tableY + 10, false, 0.7f);
        }
        if ((row + col) % 5 != 4) {
            drawEnhancedStudent(tableX, tableY - 40, false, 0.7f);
        }
    }
}

// Windows with proper frames
drawRect(windowWidth - 150, 400, 120, 180, REALISTIC_GLASS_BLUE);
setColor(Color(0.4f, 0.4f, 0.4f));
glLineWidth(3.0f);
glBegin(GL_LINE_LOOP);
glVertex2f(windowWidth - 150, 400);
glVertex2f(windowWidth - 30, 400);
glVertex2f(windowWidth - 30, 580);
glVertex2f(windowWidth - 150, 580);
glEnd();

// Additional standing students
drawEnhancedStudent(150, 200, false, 0.8f);
drawEnhancedStudent(200, 350, true, 0.7f);
drawEnhancedStudent(1050, 180, false, 0.8f);
}

// Scene 5: Enhanced Computer Lab
void drawEnhancedComputerLab() {
    // Floor and walls
    drawRect(0, 0, windowWidth, 320, Color(0.9f, 0.9f, 0.85f), true);
    drawRect(0, 320, windowWidth, windowHeight - 320, Color(0.95f, 0.95f, 0.9f), true);
    // Whiteboard with markers
    drawRect(400, 550, 400, 120, Color(0.98f, 0.98f, 0.98f));
    setColor(Color(0.3f, 0.3f, 0.3f));
    glLineWidth(3.0f);
    glBegin(GL_LINE_LOOP);
    glVertex2f(400, 550);
    glVertex2f(800, 550);
    glVertex2f(800, 670);
    glVertex2f(400, 670);
    glEnd();

    // Marker tray
    drawRect(395, 530, 410, 15, Color(0.7f, 0.7f, 0.7f));
}

```

```

// Ceiling lights
for (int i = 0; i < 5; i++) {
    drawRect(150 + i * 200, 650, 100, 15, GHIBLI_WARM_YELLOW, true);
}

// Computer desks with students
for (int row = 0; row < 5; row++) {
    for (int col = 0; col < 7; col++) {
        float deskX = 100 + col * 150;
        float deskY = 100 + row * 85;

        // Desk
        drawRect(deskX, deskY, 75, 35, Color(0.7f, 0.5f, 0.3f), true);

        // Monitor with animated screen
        drawRect(deskX + 12, deskY + 40, 45, 30, Color(0.2f, 0.2f, 0.2f), true);
        Color screenGlow = Color(0.3f + 0.1f * sin(animationProgress * 0.01f + col + row),
                                0.4f + 0.1f * sin(animationProgress * 0.012f + col),
                                0.6f + 0.05f * sin(animationProgress * 0.008f + row));
        drawRect(deskX + 15, deskY + 43, 39, 24, screenGlow, true);

        // Monitor stand
        drawRect(deskX + 32, deskY + 30, 8, 12, Color(0.5f, 0.5f, 0.5f), true);

        // Keyboard and mouse
        drawRect(deskX + 18, deskY + 3, 35, 12, Color(0.3f, 0.3f, 0.3f), true);
        drawRect(deskX + 60, deskY + 8, 6, 10, Color(0.4f, 0.4f, 0.4f), true);

        // Chair
        drawCircle(deskX + 37, deskY - 18, 12, Color(0.4f, 0.4f, 0.4f), true);

        // Students at computers
        if ((row + col) % 3 != 2) {
            drawEnhancedStudent(deskX + 37, deskY + 15, false, 0.6f);
        }
    }
}

// Teacher at front desk
drawRect(350, 480, 120, 60, Color(0.6f, 0.4f, 0.2f), true);
drawTeacher(410, 540, true, 0.8f);

```

```
}
```

```
// Scene 6: Enhanced Seminar Hall - faculty seating beside podium horizontally
```

```
void drawEnhancedSeminarHall() {
```

```
    // Floor and walls
```

```
    drawRect(0, 0, windowWidth, 260, Color(0.85f, 0.85f, 0.8f), true);
```

```
    drawRect(0, 260, windowWidth, windowHeight - 260, Color(0.95f, 0.95f, 0.9f), true);
```

```
    // Lighting
```

```
    for (int i = 0; i < 4; i++) {
```

```
        float lightX = 200 + i * 200;
```

```
        drawRect(lightX, 650, 150, 20, GHIBLI_WARM_YELLOW, true);
```

```
    }
```

```
    // Student seating - 8 rows of 10 chairs
```

```
    for (int row = 0; row < 8; row++) {
```

```
        for (int col = 0; col < 10; col++) {
```

```
            float chairX = 300 + col * 70;
```

```
            float chairY = 60 + row * 35;
```

```
            // Chair
```

```
            drawRect(chairX, chairY, 22, 12, Color(0.2f, 0.2f, 0.2f), true);
```

```
            drawRect(chairX, chairY + 12, 22, 18, Color(0.2f, 0.2f, 0.2f), true);
```

```
            // Students in chairs
```

```
            if ((row * col + row + col) % 5 != 0) {
```

```
                drawEnhancedStudent(chairX + 11, chairY + 35, false, 0.5f);
```

```
            }
```

```
        }
```

```
    }
```

```
    // Projector screen (modern classroom)
```

```
    drawRect(400, 500, 400, 240, Color(0.95f, 0.95f, 0.95f));
```

```
    setColor(Color(0.3f, 0.3f, 0.3f));
```

```
    glLineWidth(4.0f);
```

```
    glBegin(GL_LINE_LOOP);
```

```
    glVertex2f(400, 500);
```

```
    glVertex2f(800, 500);
```

```
    glVertex2f(800, 740);
```

```
    glVertex2f(400, 740);
```

```
    glEnd();
```

```
    // Projector
```

```
    drawRect(580, 450, 40, 25, Color(0.2f, 0.2f, 0.2f), true);
```

```

// Podium
drawRect(580, 420, 40, 70, Color(0.6f, 0.4f, 0.2f), true);

// Guest speaker at podium
drawTeacher(600, 490, true, 0.9f);

// Faculty seating area - horizontally beside podium (left and right)
// Left side faculty seating
for (int i = 0; i < 3; i++) {
    float facX = 350 + i * 50;
    float facY = 450;

    // Faculty chairs (different from student chairs)
    drawRect(facX, facY, 35, 15, Color(0.4f, 0.2f, 0.1f), true);
    drawRect(facX, facY + 15, 35, 25, Color(0.4f, 0.2f, 0.1f), true);

    // Faculty members sitting
    if (i < 3) {
        drawTeacher(facX + 17, facY + 45, false, 0.7f);
    }
}

// Right side faculty seating
for (int i = 0; i < 3; i++) {
    float facX = 700 + i * 50;
    float facY = 450;

    // Faculty chairs
    drawRect(facX, facY, 35, 15, Color(0.4f, 0.2f, 0.1f), true);
    drawRect(facX, facY + 15, 35, 25, Color(0.4f, 0.2f, 0.1f), true);

    // Faculty members sitting
    if (i < 3) {
        drawTeacher(facX + 17, facY + 45, false, 0.7f);
    }
}

// AC units
drawRect(windowWidth - 130, 600, 80, 40, Color(0.9f, 0.9f, 0.9f), true);
}

// Scene 7: Enhanced Library
void drawEnhancedLibrary() {

```

```

// Floor and walls
drawRect(0, 0, windowWidth, 380, Color(0.7f, 0.7f, 0.7f), true);
drawRect(0, 380, windowWidth, windowHeight - 380, Color(0.95f, 0.95f, 0.9f), true);

// Soft ceiling lighting
for (int i = 0; i < 4; i++) {
    drawRect(200 + i * 200, 680, 150, 12, GIBLI_WARM_YELLOW, true);
}

// Left wall bookshelves
for (int i = 0; i < 3; i++) {
    float shelfY = 220 + i * 160;

    drawRect(50, shelfY, 120, 150, Color(0.6f, 0.3f, 0.1f), true);

    for (int shelf = 0; shelf < 4; shelf++) {
        drawRect(55, shelfY + 15 + shelf * 35, 110, 4, Color(0.5f, 0.25f, 0.05f), true);

        // Colorful books
        for (int book = 0; book < 12; book++) {
            float bookX = 60 + book * 8;
            float bookY = shelfY + 18 + shelf * 35;

            Color bookColors[] = {
                GIBLI_SOFT_RED, GIBLI_TREE_GREEN, GIBLI_WARM_YELLOW,
                Color(0.8f, 0.4f, 0.8f), GIBLI_MINT, Color(0.8f, 0.6f, 0.2f)
            };
            drawRect(bookX, bookY, 6, 28, bookColors[book % 6], true);
        }
    }
}

// Right wall bookshelves
for (int i = 0; i < 3; i++) {
    float shelfX = windowWidth - 170;
    float shelfY = 220 + i * 160;

    drawRect(shelfX, shelfY, 120, 150, Color(0.6f, 0.3f, 0.1f), true);

    for (int shelf = 0; shelf < 4; shelf++) {
        drawRect(shelfX + 5, shelfY + 15 + shelf * 35, 110, 4, Color(0.5f, 0.25f, 0.05f), true);

        for (int book = 0; book < 12; book++) {
            float bookX = shelfX + 10 + book * 8;

```

```

float bookY = shelfY + 18 + shelf * 35;

Color bookColors[] = {
    Color(0.8f, 0.4f, 0.1f), Color(0.3f, 0.7f, 0.5f), GIBLI_LAVENDER,
    GIBLI_PEACH, Color(0.4f, 0.6f, 0.9f), Color(0.7f, 0.5f, 0.4f)
};
drawRect(bookX, bookY, 6, 28, bookColors[book % 6], true);
}
}
}

// Study tables with students
for (int i = 0; i < 4; i++) {
    for (int j = 0; j < 2; j++) {
        float tableX = 250 + i * 180;
        float tableY = 150 + j * 180;

        // Table
        drawRect(tableX, tableY, 140, 70, Color(0.6f, 0.3f, 0.1f), true);

        // Orange chairs with students
        Color chairColor = GIBLI_WARM_YELLOW;

        drawRect(tableX - 25, tableY + 10, 20, 20, chairColor, true);
        drawEnhancedStudent(tableX - 15, tableY + 35, false, 0.65f);

        drawRect(tableX + 145, tableY + 10, 20, 20, chairColor, true);
        drawEnhancedStudent(tableX + 155, tableY + 35, false, 0.65f);

        if ((i + j) % 3 != 0) {
            drawRect(tableX - 25, tableY + 40, 20, 20, chairColor, true);
            drawEnhancedStudent(tableX - 15, tableY + 65, false, 0.65f);
        }

        if ((i + j) % 4 != 0) {
            drawRect(tableX + 145, tableY + 40, 20, 20, chairColor, true);
            drawEnhancedStudent(tableX + 155, tableY + 65, false, 0.65f);
        }
    }
}

// Library officer desk
drawRect(200, 450, 100, 60, Color(0.5f, 0.3f, 0.1f), true);
drawTeacher(250, 510, false, 0.8f); // Library officer sitting

```

```

// Globe with character
drawCircle(250, 480, 20, Color(0.3f, 0.6f, 0.8f), true);
drawRect(250, 480, 10, 15, Color(0.4f, 0.2f, 0.1f), true);

// Continent details on globe
setColor(GHIBLI_TREE_GREEN);
drawCircle(250, 480, 8, GHIBLI_TREE_GREEN);
drawCircle(250, 480, 6, GHIBLI_TREE_GREEN);

// Students reading beside shelves
drawEnhancedStudent(185, 300, false, 0.7f);
drawEnhancedStudent(windowWidth - 205, 320, false, 0.7f);
drawEnhancedStudent(windowWidth - 210, 540, false, 0.65f);
}

// Scene 8: Return to Jamalkhan Bus Stop
void drawReturnToJamalkhan() {
    drawEnhancedSky();

    // Similar to Scene 0 but with evening ambiance
    drawRect(0, 0, windowWidth, windowHeight, GROUND_COLOR, true);

    // Main roads
    drawRect(0, 350, windowWidth, 100, GHIBLI_ROAD, true);
    drawRect(550, 0, 100, windowHeight, GHIBLI_ROAD, true);

    // Road markings
    setColor(GHIBLI_WHITE);
    for (int i = 0; i < 12; i++) {
        drawRect(i * 110 + 30, 390, 70, 20, GHIBLI_WHITE);
    }
    for (int i = 0; i < 8; i++) {
        drawRect(590, i * 100 + 30, 20, 60, GHIBLI_WHITE);
    }

    // Central roundabout
    drawCircle(600, 400, 45, Color(0.6f, 0.5f, 0.3f), true);
    drawCircle(600, 400, 35, GHIBLI_TREE_GREEN);

    // Monument in roundabout
    drawRect(590, 385, 20, 35, GHIBLI_CREAM, true);

    // Buildings with evening lights
    drawEnhancedBuilding(50, 450, 120, 120, GHIBLI_CREAM, Color(1.0f, 0.9f, 0.5f), 3);

```

```
drawEnhancedBuilding(200, 470, 100, 100, GHIBLI_WARM_PINK, Color(1.0f, 0.9f, 0.5f), 2);
drawEnhancedBuilding(720, 450, 130, 130, Color(0.8f, 0.9f, 0.8f), Color(1.0f, 0.9f, 0.5f), 3);
```

```
// Buildings on all four sides - properly placed on sidewalks
```

```
// Top side buildings
```

```
drawEnhancedBuilding(50, 450, 120, 120, GHIBLI_CREAM, GHIBLI_WARM_YELLOW, 3);
drawEnhancedBuilding(200, 470, 100, 100, GHIBLI_WARM_PINK, GHIBLI_WARM_YELLOW, 2);
drawEnhancedBuilding(320, 460, 140, 110, GHIBLI_LAVENDER, GHIBLI_WARM_YELLOW, 3);
drawEnhancedBuilding(720, 450, 130, 130, Color(0.8f, 0.9f, 0.8f), GHIBLI_WARM_YELLOW, 3);
drawEnhancedBuilding(880, 470, 110, 100, GHIBLI_PEACH, GHIBLI_WARM_YELLOW, 2);
drawEnhancedBuilding(1020, 460, 150, 120, GHIBLI_MINT, GHIBLI_WARM_YELLOW, 3);
```

```
// Bottom side buildings
```

```
drawEnhancedBuilding(80, 180, 140, 140, Color(0.9f, 0.8f, 0.7f), GHIBLI_WARM_YELLOW, 3);
drawEnhancedBuilding(250, 200, 110, 120, GHIBLI_SOFT_BLUE, GHIBLI_WARM_YELLOW, 2);
drawEnhancedBuilding(380, 190, 120, 130, Color(0.8f, 0.8f, 0.9f), GHIBLI_WARM_YELLOW, 3);
drawEnhancedBuilding(750, 180, 120, 140, GHIBLI_WARM_PINK, GHIBLI_WARM_YELLOW, 3);
drawEnhancedBuilding(900, 200, 130, 120, Color(0.9f, 0.9f, 0.8f), GHIBLI_WARM_YELLOW, 2);
drawEnhancedBuilding(1060, 190, 120, 130, GHIBLI_LAVENDER, GHIBLI_WARM_YELLOW, 3);
```

```
// Left side buildings
```

```
drawEnhancedBuilding(50, 580, 100, 120, Color(0.8f, 0.7f, 0.9f), GHIBLI_WARM_YELLOW, 3);
drawEnhancedBuilding(180, 600, 120, 100, GHIBLI_PEACH, GHIBLI_WARM_YELLOW, 2);
drawEnhancedBuilding(330, 590, 110, 110, Color(0.7f, 0.8f, 0.8f), GHIBLI_WARM_YELLOW, 2);
```

```
// Right side buildings
```



```

    drawEnhancedBuilding(750, 580, 110, 120, Color(0.9f, 0.8f, 0.8f),
GHIBLI_WARM_YELLOW, 3);
    drawEnhancedBuilding(890, 600, 100, 100, GHIBLI_MINT, GHIBLI_WARM_YELLOW,
2);
    drawEnhancedBuilding(1020, 590, 130, 110, Color(0.8f, 0.9f, 0.7f),
GHIBLI_WARM_YELLOW, 2);
    // Church - made more visible
    drawChurch(450, 40, 0.8f);

    // Shops on sidewalks
    drawShop(480, 500, 70, 55, Color(0.7f, 0.9f, 0.6f), "general");
    drawShop(680, 120, 75, 65, Color(0.8f, 0.6f, 0.8f), "general");

    // Trees
    drawEnhancedTree(480, 520, 1.2f);
    drawEnhancedTree(720, 540, 1.1f);
    drawEnhancedTree(750, 80, 1.0f);

    // Evening traffic
    float eveningCar1X = fmod(animationProgress * 0.35f, windowWidth + 250) - 125;
    if (eveningCar1X > -100 && eveningCar1X < windowWidth) {
        drawEnhancedCar(eveningCar1X, 370, Color(0.6f, 0.6f, 0.8f));
    }

    // Traffic on roads
    float car1X = fmod(animationProgress * 0.4f, windowWidth + 300) - 150;
    float car2X = fmod(animationProgress * 0.3f + 400, windowWidth + 300) - 150;

    if (car1X > -120 && car1X < windowWidth) {
        drawEnhancedCar(car1X, 370, GHIBLI_SOFT_RED);
    }
    if (car2X > -120 && car2X < windowWidth) {
        drawEnhancedCar(car2X, 370, Color(0.2f, 0.6f, 0.8f));
    }

    // Bus returning
    float returnBusX = windowWidth + 100 - animationProgress * 0.25f;
    if (returnBusX < windowWidth - 150) returnBusX = windowWidth - 150;
    drawEnhancedBus(returnBusX, 320, 0.9f);

    // Student walking towards bus
    if (animationProgress > 2000) {
        float studentReturnX = returnBusX - 50 - (animationProgress - 2000) * 0.03f;
        drawEnhancedStudent(studentReturnX, 380, true, 1.0f, -1);
    }

```

```
}
```

```
// Guards
```

```
drawGuard(510, 380, 0.9f);
```

```
drawGuard(680, 380, 0.85f);
```

```
// Evening ambiance overlay
```

```
setColor(Color(0.8f, 0.7f, 0.5f, 0.1f));
```

```
drawRect(0, 0, windowWidth, windowHeight, Color(0.8f, 0.7f, 0.5f, 0.1f));
```

```
}
```

Appendix B – Model Blueprints

